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Executive Summary – Retail E-Commerce Business Performance Analysis

This executive summary provides a consolidated view of business performance, profitability risks, and strategic growth opportunities identified through end-to-end data analysis of retail e-commerce operations.

1. Business Overview and Objectives

This analysis was conducted on a multi-source retail e-commerce dataset consisting of customer, product, transaction, and location-level data. The primary objective of this project was to evaluate overall business performance, customer behavior patterns, and operational risk areas using integrated relational datasets.

The key business goals of this analysis included:

- Calculating core performance KPIs such as Revenue, Profit, Return Rate, and Customer Lifetime Value (CLV)
- Identifying key revenue drivers and loss-generating segments
- Evaluating operational risks impacting profitability and customer experience
- Providing data-driven strategic recommendations for sustainable business growth

2. Data Preparation and Quality Improvement

The project involved loading and integrating four independent data tables. Significant effort was invested in data cleaning to ensure reliable and accurate downstream analysis.

Approximately 15% of customer income values were missing. Since income is a continuous numeric variable and typically right-skewed, median imputation was used instead of mean. This approach ensured that extreme high-income values did not distort the distribution while preserving all customer records. Proper handling of income data was critical to ensure accuracy in customer segmentation and CLV analysis.

Transaction date fields were converted into datetime format to enable accurate time-based trend analysis and prevent chronological sorting errors. Duplicate transaction records were identified and removed to prevent inflated revenue, order count, and customer activity metrics. Without duplicate removal, business KPIs could have been significantly overstated, leading to incorrect demand planning and inventory decisions.

The unit price field originally contained currency symbols and formatting characters, preventing numerical calculations. Regex-based cleaning was applied to remove currency symbols and commas, followed by conversion to float. This ensured accurate revenue and profit computation and prevented financial reporting errors.

Additionally, approximately 8.3% of delivery duration values were negative, which is logically impossible. Since actual delivery timestamps were unavailable for recalculation, these values were treated as data entry errors and replaced using median delivery time. This correction ensured realistic logistics performance measurement and customer experience evaluation. After cleaning, validation checks confirmed removal of missing values, correction of invalid entries, and readiness of datasets for merging and KPI computation.

3.Data Integration and Feature Engineering

All datasets were merged into a unified analytical dataset using the transaction table as the primary fact table. Left join strategy was intentionally used to preserve all transaction records and avoid revenue loss due to unmatched dimension data. Post-merge validation confirmed that row counts remained consistent (~25K transactions) and no duplicate records were introduced.

Several business-critical features were engineered, including Revenue, Gross Revenue, Total Cost, Profit, Profit Margin, and Unit Margin. Revenue was calculated using quantity, unit price, and discount adjustments. Profit was derived as total revenue minus direct product cost. Operational costs such as logistics and marketing were not included due to data unavailability.

4.Key Business Performance Insights

The business generated approximately ₹24.05 Billion in revenue but incurred a net loss of nearly ₹6.00 Billion, resulting in an overall negative profit margin of -24.96%. Nearly 49.6% of transactions were executed below cost price, indicating serious pricing inefficiencies and lack of discount control mechanisms.

Revenue patterns showed significant volatility over time, indicating dependency on campaign-driven or seasonal demand spikes rather than steady baseline growth.

Category-level analysis revealed that Electronics and Fashion categories generate the highest revenue but also contribute the largest losses, indicating structural cost or pricing issues across major product segments.

Geographically, Mumbai contributes the highest share of revenue, followed by Pune and Delhi. While this indicates strong demand concentration, it also introduces geographic dependency risk. Expanding mid-tier cities could improve revenue diversification and business resilience.

Customer analysis showed extreme revenue concentration, where the top 10 customers contribute nearly 73% of total revenue. While these customers represent strong business value, dependency on a small customer base creates revenue stability risks.

Return rate analysis showed approximately 15% of transactions are returned, which is slightly higher than standard retail benchmarks. Returned transactions were significantly more loss-making compared to non-returned orders, indicating operational inefficiencies in product quality, description accuracy, or logistics.

Discount analysis revealed a clear negative relationship between discount levels and profit. While discounting supports short-term revenue generation, excessive discounting is significantly impacting long-term profitability.

5.Strategic Recommendations

Short-Term (0–3 Months):

- Implement pricing governance frameworks with minimum margin thresholds
- Strengthen quality control processes for high-return product categories
- Launch targeted retention programs for high-value customers

Long-Term (3–12 Months):

- Expand operational and marketing investment in high-performing cities
- Optimize product portfolio by renegotiating supplier costs and phasing out persistent loss-making SKUs
- Implement structured return reduction programs through better product information and logistics improvements

6.Business Outlook

The business demonstrates strong market demand and customer engagement potential. However, long-term sustainability depends on improving pricing discipline.

Addressing pricing inefficiencies and return-driven losses alone could significantly improve overall margin performance and move the business toward sustainable profitability.

7.Limitations

- City-based merging may introduce minor mapping inaccuracies
- Seasonal decomposition analysis was not included due to time constraints
- Limited demographic attributes restricted deeper customer segmentation

8.Conclusion

Overall, the business is operating in a high-revenue but structurally unprofitable model. With focused pricing optimization, customer retention strategy, operational quality improvements, and regional expansion planning, the organization has strong potential to transition toward sustainable profitability and stable long-term growth.

The findings from this analysis provide a clear roadmap for transitioning from revenue-focused growth to profitability-focused sustainable business performance.